MIL-C-9361C
20 June 1978
SUPERSEDING
MIL-C-9361F
23 August 1976

MILITARY SPECIFICATION

CONTAINER, REUSABLE, SHIPPING AND STORAGE DISASSEMBLED, NESTED EXTERNAL AIRCRAFT FUEL TANKS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers reusable, shipping and storage containers for nested external aircraft fuel tanks.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-C-490	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings
TT-E-527	Enamel, Alkyd, Lustreless
TT-I-1795	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: AFALD/PTP, Wright-Patterson AFB OH 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-8145

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

Military

MIL-P-116	Preservation - Packaging, Method of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-D-3464	Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-R-6855	Rubber, Synthetic, Sheets, Strips, Molded or Extruded Shapes
MIL-W-6858	Welding, Resistance: Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-R-7575	Resin, Polyester, Low-Pressure Laminating
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback; Cold-Application
MIL-P-16232	Phosphate Coating, Heavy, Manganese or Zinc Base (for Ferrous Metals)
MIL-P-17549	Plastic Laminates, Fibrous Glass Reinforced, Marine Structural
MIL-P-19834	Plate, Identification, Metal Foil, Adhesive Backed
MIL-P-23377	Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant
MIL-I-26860	Indicator, Humidity, Plug, Color Change
MIL-P-26915	Primer Coating, Zinc Dust Pigmented, For Steel Surfaces
MIL-V-27166	Valve, Pressure Equalizing, Gaseous Products

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

MIL-W-45205	Welding, Gas Metal-Arc and Gas Tungsten-Arc, Aluminum Alloys, Readily Weldable for Structures, Excluding Armor
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate, For Aerospace Applications

STANDARDS

Federal

FED-STD-101	Preservation, Packaging, and Packing Materials: Test Procedures
FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence For the Selection Of
MIL-STD-648	Design Criteria For Specialized Shipping Containers
MIL-STD-831	Test Reports, Preparation Of
MIL-STD-889	Dissimilar Metals
MIL-STD-1261	Welding Procedures For Constructional Steels

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Society of Automotive Engineers (Aerospace Material Specification)

AMS 3227	Synthetic Rubber, Hot Oil and Coolant Resistant, Low Swell 55-65

(Copies of SAE publications may be obtained from the Society of Automotive Engineer, Inc., Two Pennsylvania Plaza, New York NY 10001.)

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:09Z
Check the source to verify that this is the current version before use.

National Bureau of Standards

Handbook H28 Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20330.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article inspection. This specification makes provisions for first article inspection (see 4.3).

3.2 Materials. Materials shall conform to the requirements specified and specifications and drawings referenced herein or as otherwise specified by the procuring activity (see 6.2). Any deviation from the fiberglass reinforced plastic (FRP) process in 3.2.1 or metal composition requires prior approval of the procuring activity. Wood shall be limited to possible use for internal blocking and bracing (see 3.2.3). The use of magnesium is prohibited.

3.2.1 Molding. The following process which utilizes fiberglass reinforced plastic has been approved for use by the Government. The FRP container fabrication approach covered by this specification shall consist of applying alternating layers of fiberglass mat, resin, and woven roving cloth in accordance with MIL-P-17549. The resin shall conform to MIL-R-7575. The number of laminations shall be as required to provide structural strength necessary to meet the performance and test provisions of this specification. The surfaces shall have a molded-in gel coat finish to provide abrasion resistance and the specified color (see 3.7.3).

3.2.2 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MIL-STD-889 shall not be used in intimate contact with each other.

3.2.3 Protective treatment. When materials are used in the construction of containers that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3 Selection of specifications and standards. Specifications and standards for all materials specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.3.1 and 3.3.2.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:09Z
Check the source to verify that this is the current version before use.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids or request for proposal, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used, provided:

a. They can be replaced by standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.3.2 Standard parts. With exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3.3 Certification. Government certification and approval of special processes and equipment shall be as required by applicable specifications.

3.4 Design and construction. The container shall be designed to prevent deterioration resulting from environmental exposure and from mechanical shock and vibration. The container shall consist of a cover section and a base section where the base section shall always form the lower part of the container. The cover section shall be not less than 1/2 of the overall assembled container height and shall be provided with handling provisions for mechanical lifting, utilizing standard material handling equipment. The cover shall attach to the base by container closure hardware that is integral to the container.

3.4.1 Dimensions and weight. The container shall be of the minimum size and practical weight, consistent with the performance requirements of this specification.

3.4.2 Tank mounting provisions. Each container shall include complete interior supports and attachments to safely, quickly, and easily secure and release the model(s) or type(s) of tanks prescribed by the procuring activity (see 6.2).

3.4.3 Standardization. The container's internal component parts shall be so designed that they will be standard, insofar as practicable, among the various and succeeding models of the basic tank. The container design, including saddles and tiedown restraint devices, shall be configured to accept those tanks specified by the procuring activity (see 6.2).

3.4.4 Assembly and disassembly.

3.4.4.1 Closures. Closure fasteners shall be attached and integral to the container and shall be such that only common hand tools will be required to open and close the container. The assembled and closed container shall meet the requirements of this specification without the aid of separately applied sealing compounds or materials.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:09Z
Check the source to verify that this is the current version before use.

3.4.4.2 Accessibility. The container shall be designed to provide ready accessibility to interior supports and shall permit simple insertion and removal of the nested tanks by hand and/or by means of overhead lifting devices through a sequence of simple and safe operations.

3.4.4.2.1 Component access. When the container is designed for a tank containing components such as but not limited to rubber goods, which are subject to periodic inspection tests, provisions shall be made to allow access to those components without removing the container from its position in a stack of similar containers. This will require locating such components within the container adjacent to the access door. The access door to components and desiccant receptacle(s) may be the same.

3.4.4.2.2 Access door. All container accessory items (e.g., pressure equalization valves, humidity indicators, etc) shall be integral to the access door which shall be located at one end of the container and easily accessible for inspection under row storage conditions.

3.4.4.2.3 Records accessibility. The records receptacle shall be integral to the access door in such a manner as to allow access to the records without opening the container, removing the access door, and without destroying the internal preservation. In addition, the records shall be accessible by use of ordinary hand tools.

3.4.4.3 Time for removal of contents. The time required to open the container, remove the tanks, and separate the sections, shall not exceed ten man-minutes per tank for tank capacity up to 1,000 gallons (3,785 liters).

3.4.5 Tooling/mold design. The tooling/mold used to fabricate prototype containers (first articles) shall be designed to permit transition to production scale requirements. Structural, material, or design changes to the container shall not be created by the transition without approval of the procuring activity.

3.4.6 Free drainage. The container's external pockets shall be constructed to insure that in the normal storage position every possible pocket, large or small, is provided with a means of drainage. When necessary, holes shall be drilled of sufficient size that they may be adequately painted without danger of subsequent stoppage. When drainage holes are not practical, the affected area shall be filled with a filler compound of suitable consistency in such manner as to prevent the collection of water, or treated to insure no corrosive action.

3.4.7 Handling provisions. The container shall be provided with fork-lifting, hoisting, towing, and tiedown provisions commensurate with the size and maximum gross weight which are expected to be handled or constrained. These provisions shall insure safe and efficient movement by all modes of transportation. Handling provisions such as rings, eyes, lugs, etc., when not in use, shall not extend beyond the limits of maximum

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

body dimensions of the container. Lifting rings or eyes installed for handling major sections during disassembly shall be located to insure a stable lifting configuration. Rings and eyes shall have a minimum inside diameter of 2-1/2 inches (63.5 mm) for a container with gross weight of 2,200 pounds (997.9 kg) or less and a minimum inside diameter of 3 inches (76.2 mm) for gross weight over 2,200 pounds (997.9 kg). The container with nested tanks and all components in place, shall conform to the requirements herein.

3.4.7.1 Forklifting. Forklift entries shall be provided from both sides of the container. The forklift entry shall have a minimum clear height of 2-7/8 inches (73.0 mm). Width and spacing shall be in accordance with MIL-STD-648. Forklift handling of the container shall exhibit no significant instability on the tines (e.g., out of balance or tending to fall off or capsize) or tendency of the tines to permanently deform, fracture or puncture the container when tested as specified in 4.6.5.1.

3.4.7.2 Hoisting. Hoisting provisions shall, in addition to the requirements in 3.4.7, be placed as far away from the loaded center of balance as the strength of the container will safely permit. Hoisting provisions shall not create any unsafe, awkward, time consuming or potentially dangerous handling conditions, nor shall any permanent deformation of the hoisting provisions or supporting structure be observed when tested in accordance with 4.6.5.1.

3.4.7.3 Towing. Towing provisions shall, in addition to the requirements in 3.4.7, be provided on each end of the container. Towing provisions shall be capable of safely supporting the fully loaded container as a hoisting provision and shall be tested as a hoisting provision in accordance with 4.6.5.1.

3.4.7.4 Tiedown. Tiedown provisions, in addition to the requirements in 3.4.7, shall be compatible with the attachment points on aircraft cargo floors which, in general, have a capacity of 10,000 pounds (4,536 kg) and are spaced on 20 inch (508 mm) centers. There shall be a minimum of two tiedown provisions provided on each of the longest sides and a minimum of one tiedown provision on each end of containers wider than 5 feet (1.52 meters). Tiedown provisions shall not create any unsafe, awkward, time consuming or potentially dangerous handling conditions, nor shall any permanent deformation of the tiedown provisions or supporting structure be observed when tested in accordance with 4.6.5.2.

3.4.7.5 Indexing. Each container shall be provided with indexing provisions that shall encourage rapid and orderly stacking operations for like type containers. These provisions shall provide lateral and longitudinal structural restraint and shall be tested in the forklift test by stacking two containers and proceeding to repeat the test specified in paragraph 4.6.5.1 for forklifting.

3.4.8 Stacking. Means shall be provided on the top and bottom of all containers or skids to permit stacking and prevent sliding or slipping in the longitudinal and lateral horizontal plane. The container shall be capable of supporting the superimposed weight of two like containers including the contents, or a

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

number of like containers with contents stacked to a height of 16 feet (4.88 meters) whichever is greater. The loaded containers shall be capable of withstanding the dynamic tests in section 4 without any permanent deformation or creation of potentially dangerous handling conditions.

3.4.9 Service facilities. Containers shall be designed to maintain dehydrated interior conditions under conditions of tests specified herein. The following provisions shall be incorporated in each container.

3.4.9.1 Sealing. All joining surfaces shall be such as to insure sealing under performance requirements of this specification. Sealing gasket material shall conform to AMS3227 or MIL-R-6855 class 2, type B, grade 60. All gasket joints shall be vulcanized to form a continuous gasket. When assembled, the sections of the container shall be designed in such a manner that no loads will be applied to the gasket(s) other than those required for sealing. The container, including joints and gaskets, shall be designed, and constructed of materials which will provide a water vaporproof seal in accordance with Method II of MIL-P-116.

3.4.9.2 Humidity indicators. Humidity indication shall be obtained by means of a plug-type humidity indicator conforming to MIL-I-26860. The humidity indicator nut shall be tacked welded or adhered to inside of container to provide a capability for removal and replacement of indicator without removing container cover. The externally viewed humidity indicator shall be installed in the access door of each container to determine internal humidity without opening the container. The indicator shall be positioned to be easily read and to be protected from damage during normal container storage and transportation cycles.

3.4.9.3 Relief valves. Unless otherwise specified, each container shall be provided with an automatic relief valve system (see 6.2). In addition, a manually operated relief valve as specified in 3.4.9.3.1 shall be provided unless a manual feature is built into the automatic valve system. The valve(s) shall not be removable without the use of tools. A metallic decal providing operating instructions shall be applied adjacent to the valve(s). All relief valves shall be mounted on the access door of the container.

3.4.9.3.1 Manual relief valve. The valve shall be brass, aluminum or stainless steel captivated type screw having a flat machined surface along the side of the screw which will provide the necessary relief outlet. A rubber washer (conforming to MIL-R-6855) shall be placed under the head of the screw to form the seal when valve is in the closed position. Pipe threads shall be in accordance with Handbook H28, Part No. 2.

3.4.9.3.2 Automatic relief valve. The automatic relief valve(s) shall conform to MIL-V-27166, type III, class 1, unless otherwise specified (see 6.2). The valve(s) shall not leak in the flow direction at pressure differentials less than 0.5 psig (3.45 kPa) for vacuum relief and pressure relief. The valve(s) shall open at any pressure greater than 0.5 psig (3.45 kPa) that is consistent with the structure of the container.

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:09Z
Check the source to verify that this is the current version before use.

3.4.9.4 Location of service items. The access door to components requiring periodic inspection or test, if required, the plug-type humidity indicator and the relief valve shall not extend beyond extremities and shall be located on one end of the container so that the items will be easily accessible for inspection under row storage conditions.

3.4.9.5 Desiccant holder. The container shall be equipped with interior receptacle(s) for desiccants. Each receptacle shall include a refillable enclosure which shall retain bagged desiccant conforming to MIL-D-3464 while permitting the passage of air. The enclosure shall be accessible through the access door of the container. The total capacity of the receptacles shall be determined by the quantity of desiccant required in accordance with MIL-P-116.

3.4.10 Dynamic tests. The fully loaded container shall be capable of withstanding the dynamic tests in 4.6.3 without any permanent deformation or creation of potentially dangerous handling conditions.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other. Changes in the manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-1000. Major sections of the container which may be otherwise assembled in more than one relative position shall be furnished with an assembly guide(s) and suitable markings which will permit assembly in one relative position only.

3.6 Performance.

3.6.1 Leakage. The container leakage test shall be in accordance with Method 5009 of FED-STD-101 except that 1 psig (6.90 kPa) will be used as the minimum test pressure. Pressure testing will be conducted before and after the first article inspection (see 4.6.2).

3.6.2 Ability of container to protect contents. When tested in accordance with section 4, the contents of the container shall not show any indications of damage that would affect their utility. There shall be no evidence of a substantial amount of shifting of the contents within the container that would create conditions likely to cause damage during shipment. Unless otherwise specified, examination of the contents shall be performed before and after the tests by or under the supervision of the procuring activity. Examination of contents shall consist of visual inspection of tank components (see 4.6.1).

3.6.3 Objective. The containers designed and constructed as specified herein shall function as shipping and storage containers to include world-wide storage and transportation of external fuel tanks when the tanks are prepared for shipment in accordance with the preservation requirements as specified by the procuring office. The quantity of tanks to be accommodated in each container shall be as specified by the procuring activity (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:09Z
Check the source to verify that this is the current version before use.

3.7 Finish. The surface treatment for materials other than those listed below shall require approval of the procuring activity. The exterior surfaces of all containers shall be color conforming to color number 24052 of FED-STD-595.

3.7.1 Aluminum surfaces. Aluminum surfaces shall be anodized in accordance with MIL-A-8625 or conversion coated in accordance with MIL-C-5541. All aluminum surfaces shall be painted with primer in accordance with MIL-P-23377 and overcoated with material conforming to TT-E-527 or MIL-C-83286.

3.7.2 Steel surfaces. All ferrous metal surfaces shall be cleaned in accordance with TT-C-490 except that threaded or other working surfaces shall be treated in accordance with type M, class 4B of MIL-P-16232. All ferrous metal surfaces shall be painted with primer in accordance with MIL-P-26915 type I and overcoated with material conforming to TT-E-527 or to MIL-C-83286.

3.7.3 Fiberglass reinforced plastic surfaces. All external surfaces of the fiberglass reinforced plastic container shall have a neopentyl glycol (2, 2-dimenthyl-1, 3-propanediol) gel coat integrally colored 10 to 30 mil thickness. The gel coat shall be pigmented with color conforming to the requirements of 3.7.

3.7.4 Container hardware. All latches, handles, hinges, hinge pins, fasteners, bolts, eyes, and other container hardware shall be of corrosion resistant material or treated in such a manner as to prevent corrosion. The container hardware shall not be susceptible to corrosion damage which in any way impairs specified performance of the individual parts. All cadmium plating shall be class 1, type II as defined in QQ-P-416. All cadmium plated items shall be painted. Aluminum and aluminum alloy parts not otherwise protected shall be anodized in accordance with MIL-A-8625 or shall be given other surface protection as may be specifically authorized by the procuring activity. All hardware shall be mechanically attached and shall be recessed or protected to minimize handling damage.

3.8 Markings. Markings shall be of waterproof ink or paint. Ink shall conform to TT-I-1795 and paint to TT-E-527. Color of markings shall be white, conforming to color No. 37875 of FED-STD-595. The following markings shall be applied to all containers at the locations and in the sizes indicated:

a. Adjacent to the lifting rings, eyes, or lugs in 2-inch (50.8 mm) letters:

LIFT HERE

Arrows 5 inches (127.0 mm) long shall point to the rings, eyes, or lugs.

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:09Z
Check the source to verify that this is the currentversion before use.

b. On opposite sides of the upper half of the container corresponding to lift truck openings in 4-inch (101.6 mm) letters:

DO NOT DROP

DO NOT SET ON END

c. Adjacent to the plug-type humidity indicator in 1-inch (25.4 mm) letters:

HUMIDITY INDICATOR

d. At the loaded center of balance on both sides of the container, a vertical line 6 inches (152.4 mm) long and 1 inch (25.4 mm) wide with the marking adjacent in 1-inch (25.4 mm) letters:

CENTER OF BALANCE

e. On access door designate in 1-inch (25.4 mm) letters:

DESICCANT RECEPTACLE INSIDE

(Specify units of desiccant)

f. Adjacent to the relief valve in 1-inch (25.4 mm) letters:

RELIEF VALVE

g. Along both sides in 2-inch (50.8 mm) letters:

REUSABLE CONTAINER - DO NOT DESTROY

h. In 1-inch (25.4 mm) letters designate forklift entries:

FORKLIFT

i. In 1-inch (25.4 mm) letters designate in appropriate locations:

SLING LIFT POINTS

j. In 1-inch (25.4 mm) letters designate in appropriate locations:

TIEDOWN

k. Adjacent to the record receptacle in 1-inch (25.4 mm) letters:

RECORDS/INSTRUCTIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

l. On forward end of containers both top and bottom shall be identified in 1-inch (25.4 mm) letters to insure proper mating of the two sections:

FORWARD END

m. On one side in 2-inch (50.8 mm) letters:

TO OPEN CONTAINER

n. Followed in 1-inch (25.4 mm) letters by the specific instructions in detail for opening the container, lifting the cover section, and any other details required for opening the container.

o. Adjacent to each cover lifting device in 2-inch (50.8 mm) letters:

COVER LIFT ONLY

Arrows 5 inches (127.0 mm) long shall point to the cover lifting devices.

3.9 Identification of product. The containers shall be marked for identification in accordance with MIL-STD-130. The name plate shall be in accordance with type II of MIL-P-19834.

3.10 Welding. Welding shall be in accordance with MIL-STD-1261, MIL-W-45205, or MIL-W-6858. Welds shall be reasonably smooth and free of craters and porosity. They shall exhibit characteristics of fusion, penetration, and soundness of weld deposit representative of good welding practice. All welding fluxes, scale, weld spatter, acids, or basic solutions shall be completely removed prior to application of any finish coats.

3.11 Workmanship. The container shall be of clean design, well made, and free from any defects which may affect durability, strength, and serviceability.

3.12 Instruction document. An instruction document shall be provided to describe the procedure for packing and unpacking tank components. The document shall include the necessary details, illustrations, parts list, and procedure for opening and closing the container. The instructions shall be stowed in the record receptacle and shall accompany each container. The instructions shall also describe materials needed, procedures for field maintenance of the containers, and instructions for proper blocking and bracing. A table listing the necessary restraint data developed in 3.13 shall be included.

3.13 Restraint requirements. Data shall be provided in the instruction book (see 3.12) for restraining containers subjected to a maximum steady wind of 78 knots (40.1 meters per second). Data shall be provided for each container increment up to 16 feet (4.88 meters) in height (see 4.3.3).

Source: https://assist.dla.mil -- Downloaded: 2026l0-17T12:09Z
Check the source to verify that this is the currentversion before use.

3.14 Records receptacle. An externally accessible records receptacle shall be provided on the access door of the container for records, instructions, and related documents. The receptacle shall be cylindrical or rectangular. Cylindrical receptacles shall have minimum inside dimensions of 1.7 inches (43.2 mm) in diameter by 9 inches (229 mm) in length. Rectangular receptacles shall have minimum inside dimensions of 2 inches (50.8 mm) by 9 inches (229 mm) by 12 inches (305 mm). The closure for the receptacle shall be permanently attached and capable of being removed and replaced by hand or common hand tools. The receptacle and its closure shall be designed to protect its contents from water and physical damage and to permit ready access.

3.15 Container stacking on military pallets. Containers with payloads stacked on military pallets with cargo nets in place shall be compatible with the usable cargo space for C-130 and C-141 aircraft.

3.16 Special environmental considerations. Container storage and use shall be under any temperature conditions between -40°F (-40°C) and +140°F (+60°C), under all relative humidities and in steady winds to 78 knots (40.1 meters per second) when properly anchored. In addition, the containers shall be capable of sustaining a snow load of 40 pounds per square foot (195 kg/m^2) of horizontal surface area.

3.17 Maintainability. All containers shall be maintenance free up to five years of operational life or ten years of storage life except for minor repairs that may be caused by accidental handling damage. All repairs of the containers shall be capable of being accomplished in the field without use of special tools. Meantime to repair (MTR) such minor damage shall not exceed 1.5 hours per task. Corrective maintenance procedures shall be furnished by the supplier (reference para 3.12).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection and testing of nested, external aircraft fuel tank shipping containers shall be classified as follows:

a. First article inspection (4.3)

(1) Preliminary first article inspection. For the purpose of this specification, preliminary first article inspection shall consist of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

those tests performed on sample containers to assure adequacy and effectiveness of the container engineering design and to insure preparedness for the conducting of first article test.

(2) Final first article inspection. For the purpose of this specification, final first article inspection shall consist of those tests performed on sample containers, conducted at the tank manufacturer's plant or container manufacturer's plant and witnessed by a technical representative of the procuring activity to assure ability of the container to conform to this specification.

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. One container, representative of production design and construction, shall be subjected to a preliminary first article inspection consisting of all inspections specified under 4.6. After successful completion of the preliminary inspection, the container, or one identical thereto, shall be subjected to the tests under 4.6 for final first article inspection (see 4.3.2).

4.3.1 Location of first article inspection. Preliminary first article inspection of the container shall be performed at the container manufacturer's plant. Final first article inspection shall be performed at the tank manufacturer's plant or container manufacturer's plant (see 6.2).

4.3.2 Requests for first article inspection. Two weeks prior to the date upon which final first article inspection is to begin, the contractor shall communicate with the procuring activity in order that the procuring activity may send a technical representative to witness the inspection. The request shall be accompanied by a preliminary first article test report which shall include a detailed description of the examination and tests performed, the procedures used, and the results. Interior and external photographs, a copy of assembly drawing(s) of the container, and the packaging pattern shall accompany the report.

4.3.3 First article test report. After the contractor completes the final first article inspection, he shall prepare a first article test report in accordance with MIL-STD-831. Restraint data analysis, photographs, and assembly drawings of the container in accordance with MIL-D-1000 shall be included in the report.

4.4 Quality conformance inspection. Each container submitted for acceptance shall be subjected to and shall pass the examination of product (see 4.6.1).

4.5 Test conditions. For both preliminary and final first article inspection, tests shall be performed with the container fully assembled and prepared for shipment.

4.5.1 For container only. The test load used in testing the design and fabrication of the container may be reparable items for which the container

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:09Z
Check the source to verify that this is the current version before use.

was designed or a simulated load of equal weight and mounted in the same manner and location within the container.

4.6 Inspection methods.

4.6.1 Examination of product. Examination of product shall be for defects classified in Table I. Any major defect found shall be cause for rejection of the item.

TABLE I. Table of defects.

Examine	Defects	Category Major	Category Minor
Material	Material not type as specified	X	
Construction	(a) Any component missing or malformed	X	
	(b) Any detail or feature of fabrication not as specified on drawings.	X	
Workmanship	Containers manufactured shall be free from imperfections that may affect their utility.	X	
Stacking Provisions, Rings, Eyes, Lugs, Indexes	Not as specified.	X	
Finish	Ferrous/Non-ferrous metals and non-metals not treated, primed, painted as specified.	X	
Markings	Omitted, incorrect, illegible, of improper size, location, imcomplete		X
Welding and fasteners	(a) Welds not sound, smooth, or free of craters.	X	
	(b) Welds not clean of flux, scale, spatter.		X
	(c) Rivet heads not tight, free of cracks.	X	
Interchangeability	All parts of the same model will be completely interchangeable with other containers.	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

Examine	Defects	Category Major	Minor
Color	Not as specified.	X	
Drainage	Pockets on the containers collect water.	X	
Record Receptacle	Not as specified.		X
Instruction Document	Not as specified.		X
Tank Component	Damages that exceed those specified in applicable technical manual.	X	

4.6.2 Leakage. The container shall be examined for leakage when assembled, loaded, and sealed in the same manner it will be prepared for shipment. The pressure relief valve(s) shall be removed and the hole sealed. The container shall be tested as specified in Federal Test Method Standard No. 101, method 5009, and the minimum pressure shall be 1 psig (6.89 kPa). The leakage test of the loaded container shall be performed prior to and after the tests specified in 4.6.3 through 4.6.5.2. The purpose of the container leak test is to determine the ability of the closure and seal design to provide a method IId or IIf, as applicable, water-vaporproof barrier environment.

4.6.2.1 Preferred method. The container shall be immersed slowly into water until the closure flange is approximately 4 inches (101.6 mm) below the surface. The container shall then be examined for leakage. During the examination, the water shall be quiet and free of bubbles. Following immersion, any small closure or welds which were not under water shall be tested for leaks by the application of a soap solution.

4.6.2.2 Variant to preferred method. As an alternate to the above method, the container shall be tested for leakage by applying a soap solution to all seams and closures of the pressurized container.

4.6.3 Dynamic tests. The container shall be loaded and subjected to the following tests in accordance with Federal Test Method Standard No. 101 (see 3.4.10). Test a, b, and c shall be subject to temperature extremes from -20°F to 140°F (-28.9°C to 60.0°C).

a. Edgewise drop (rotational) test, Method 5008

b. Cornerwise drop (rotational) test, Method 5005

c. Impact test, Method 5023 or Method 5012

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

d. Vibration (repetitive shock) test, Method 5019

e. Towing test, Section 6.6 of Method 5011

f. Pushing test, Section 6.5 of Method 5011

4.6.4 Stacking strength test. The container shall be tested in accordance with Method 5016 of Federal Test Method Standard No. 101. The container shall be considered as a level A package with a superimposed load calculated by the following formula:

$$W = P \times \frac{16-H}{H} \times S$$

In addition to this test, if the principal support structure is a non-metallic material, the test shall be repeated at a temperature of 120°F ± 5°F (49°C ± 3°C) and 90 percent relative humidity for a period of 168 hours where the load is calculated with the constant "S" as 1.0.

4.6.5 Mechanical handling test. The container shall be loaded and subjected to the tests specified herein.

4.6.5.1 Forklifting, hoisting, and towing. The container shall be tested in accordance with the appropriate section of Method 5011 of Federal Test Method Standard No. 101 (see 3.4.7.1, 3.4.7.2, 3.4.7.3, and 3.4.7.5).

4.6.5.2 Tiedown strength test. Each tiedown provision shall be identified for test purposes. A load shall be applied to each tiedown provision in the same way that it would be applied in service. In the absence of clearly specified tiedown procedures, the load shall be applied at an angle of 45° downward from the horizontal and simultaneously 45° outboard from the container surface. The test load shall be equal to that which would be applied in accordance with 3.3.4 of MIL-A-8421. The load shall be applied in each of the directions which may reasonably result during active shipment. The duration of the test shall be one minute in each of the test directions (see 3.4.7.4).

4.7 Drainage. Water shall be placed over the entire closed container. Visual observation shall be recorded and included in first article test report (see 3.4.6).

4.8 Rejection and retest criteria. Failure of any type of container to conform to any of the requirements of this specification shall be cause for rejection of that type of container. If the container fails in any individual inspection, demonstration, or test, it shall be replaced by an operable unit and reverified. If the verification container then qualifies and an engineering analysis of the failure indicates that the original failure was not induced by design, the container shall have met the requirements of this specification. The analysis shall be available to the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

4.9 Packaging. Packaging shall be inspected for conformance to section 5.

5. PACKAGING

5.1 Preparation. Unless otherwise specified, each container shall be completely assembled for shipment. Tank mounting plates or brackets shall be mounted in place. Any other accessories shall be securely fixed at any convenient place on the interior of the container. Closure bolts or other closure devices shall be assembled and drawn to apply a firm but not excessive pressure. All exposed operating surfaces of closure devices and exposed threaded surfaces shall be treated with an application of material conforming to grade 1 of MIL-C-16173.

5.2 Marking of shipments. The container shall be marked for shipment in accordance with MIL-STD-129. The shipment marking shall include the following:

CONTAINER, REUSABLE, SHIPPING AND STORAGE,
DISASSEMBLED, NESTED EXTERNAL AIRCRAFT FUEL TANKS
SPECIFICATION MIL-C-9361G

5.3 Preservation-packaging. Preservation-packaging shall be in accordance with MIL-P-116, Method IId or Method IIf if shipped with fuel tanks enclosed, all other shipments shall be Method III.

6. NOTES

6.1 Intended use. The containers covered by this specification are intended for domestic and overseas use as shipping and storage containers for nested external aircraft fuel tanks.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. The model number(s) or type(s) of the fuel tank(s) to be accommodated (see 3.4.3 and 3.4.2).

c. The applicable drawing(s).

d. The number of containers to be furnished.

e. Where the first article inspection samples should be sent, the activity responsible for inspection, instructions concerning the submittal of the test report (see 4.3.1 and 4.3.2), and activity responsible for supplying necessary fuel tanks.

f. Whether an automatic relief valve system should be furnished with the container (see 3.4.9.3).

g. Type of relief valve if different than specified (see 3.4.9.3.2).

Source: https://assist.dla.mil -- Downloaded: 202610-17T12:09Z
Check the source to verify that this is the current version before use.

h. Whether special stacking, storage, and storage provisions are required as well as any additional markings.

i. Material required if different from those in this specification or drawings (see 3.2).

j. Quantity of tanks to be accommodated (see 3.6.3).

6.3 Contract data requirements. All data requirements necessary for design reviews and evaluation and for completion of first article test reports shall be listed on a DD Form 1423 in the contract.

6.4 First article. Invitation for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Custodian:
Navy - AS
Air Force - 69

Review activities:
Air Force - 11, 99

User activity:
Air Force - 99

Preparing activity:
Air Force - 69

Project No. 8145-0003

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:09Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-9361 CONTAINER, REUSABLE, SHIPPING AND STORAGE, DISASSEMBLED, NESTED EXTERNAL AIRCRAFT FUEL TANKS

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 PREVIOUS EDITION WILL BE USED.

Source: https://assist.dla.mil -- Downloaded: 202640-17T12:09Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DEPARTMENT OF AIR FORCE
DOD-318



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

AFALD/PTP
Wright-Patterson AFB OH 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T12:09Z
Check the source to verify that this is the current version before use.